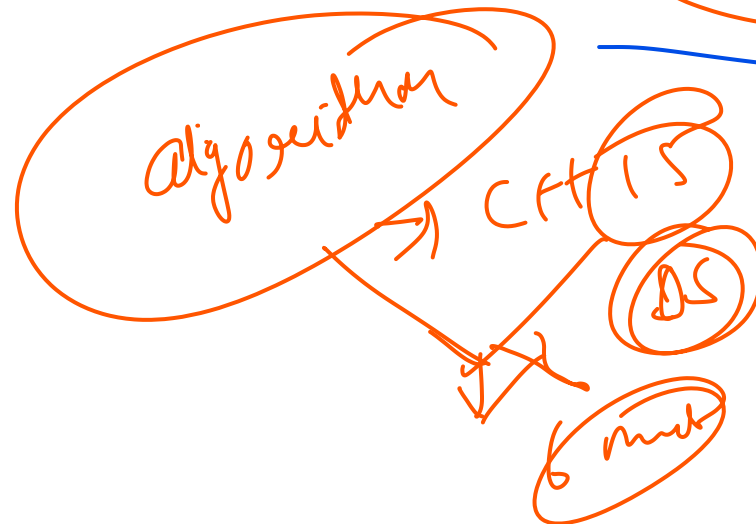


$O(n^2) \Rightarrow WC$
 $O(n \log n) \Rightarrow BC, AC$

Lec_42

Quick Sort: Time Complexity Analysis

In-depth



```

qs(a, p, q)
{
  if (p == q) return a[p];
  m = partition(a, p, q)
  qs(a, p, m-1)
  qs(a, m+1, q)
  return a;
}

```

m = position of pivot

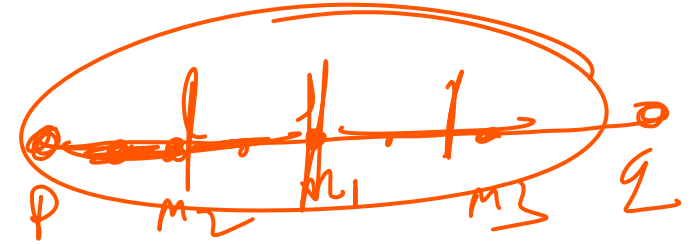
$$T(n) = c + n +$$



For best and Avg Case: Dividing Almost in Equal parts

Ex1:

$$T(n) = c + n + T(n/2) + T(n/2) + c$$



$$T(n) = 2T(n/2) + n + 2c$$

$$T(n) = 2T(n/2) + n$$

$$T_c = O(n \log n)$$

$$T_c = O(n \log n)$$

$$f(n) = n^{\log_b a}$$

$$n^{\log_2 2}$$

$$n \log n$$

$$n \log n$$

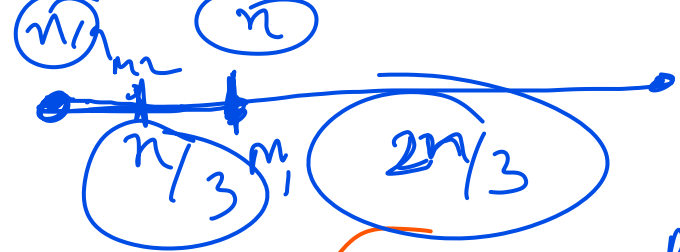
$$O(n \log n)$$

when array is not sorted

already

Ex2:

if pivot is coming at $n/3$ index of the array then



$$T(n) = \cancel{O(n)} + T(n/3) + T(2n/3) + C$$

$$T(n) = T(n/3) + T(2n/3) + n$$

RTM

Q(L, P, R)
if —
m = part(L)
Q(L, P, m-1)
Q(L, m+1, R)
return

$$TC = n \times \log_3 n$$

$$TC = n \log_3 n$$

$$TC = n \log n$$

$$2n = 3^{n_2}$$

$$n/3 \Rightarrow n/9$$

$$n/3 \Rightarrow 2n/9$$

$$\frac{2n}{3^{n_2}} = 1$$

$$T(n) = T(n/3) + T(2n/3) + n$$

$$T(n/3) = T(n/9) + T(2n/9) + n/3$$

$$\frac{n}{3^{n_1}} = 1$$

$$n = 3^{n_1}$$

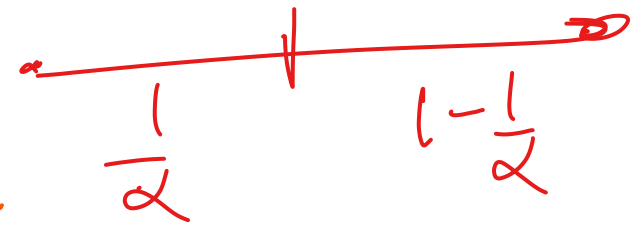
$$n_1 = \log_3 n$$

$$n_2 = \log_2 (2n)$$

$$n_2 = \log_2 n + \log_2 2$$

Ex3:

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n-1}{2}\right) + n$$



General
Case

RTM

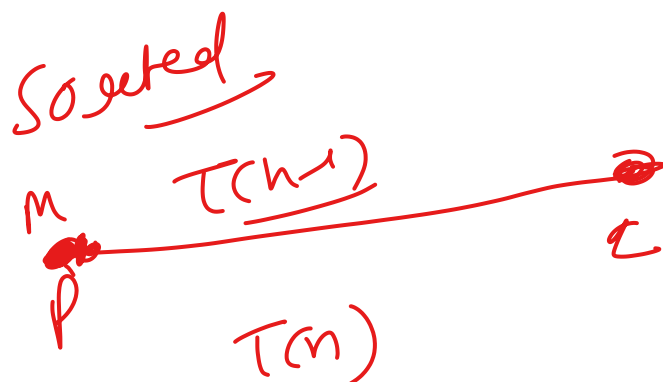
In general:

For Worst Case: One Sided Division

if pivot is already in its correct place

means array is already sorted

$T_c = O(n)$



$T(n) = T(n-1) + T(1) + n$

$T(n) = T(n-1) + n$

$T_c = O(n^2)$

→ In Best Case:
Avg Case

$$na \Rightarrow T(n) = 2T(n/2) + n \\ \Downarrow \\ O(n \log n)$$

space
 $O(\log n)$

→ Worst Case:

$$\Rightarrow T(n) = T(n-1) + n \\ \Downarrow \\ TC = O(n^2)$$

space
 $O(n)$

→ Inplace,

not stable
algorithm

Imp points:

Sorted $\rightarrow O(n^4)$

- Quick sort is useful for Completely unsorted array gives $(n \log n)$
- If Array is already sorted quick sort takes $O(N^2)$, So quick sort is not preferred for Sorted or almost sorted array, But In this case insertion sort takes $O(n)$.

if array is already sorted or almost sorted
qs $= O(n^2)$
ss $\rightarrow O(n)$

$\Rightarrow$ practice qn of quick sort

Thank you

